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Reduced Feature Space for Detection of Diabetes Mellitus Type 2

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Abstract
This study had two main objectives. The first was to investigate the possibility of formulating a minimal feature space based on information measures taken from continuous interstitial glucose monitor records to detect the presence of diabetes mellitus type 2. Then, use the constructed feature space to be used in a machine learning algorithm to develop a model capable to detect records with the pathology and without. The parameters forming the feature space were also intended to characterise the condition, thereby advancing the understanding of its dynamics. The results showed that the entropy-complexity plane, modified using weighted permutation entropy, was a satisfactory choice for defining the feature space. This was complemented by applying a deterministic k-means algorithm to develop the final detection model. Furthermore, a method is proposed for combining one-dimensional signal databases from different sources to create more comprehensive ones. The results obtained contribute to the informational and algorithmic knowledege and found aspects to assist to understand the dynamics the pathology.

Keywords Permutation Entropy, Type 2 Diabetes, Shannon Entropy, Complexity, k-means algorithm

AMS 2010 subject classifications 94A17, 92C55

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1. Introduction

1.1. Type 2 diabetes

Type 2 diabetes mellitus (T2DM) is one of the most significant health problems in Latin American coutries [1]. Just in Argentina, 4.3 million people (9% of the population of the country) were dealing with diabetes in 2024, with a forecast of 5.9 millions patients by 2050. Worldide, more than 422 millions people are affected by the disease and the number is still growing.

This metabolic deficiency is characterized by poor glucose regulation, extreme tissue resistance, and incorrect hormone dosing, leading to severe health deterioration over time if not strictly controlled. Experts have observed in the last decade a growing prevalence of prediabets, that can transform to actual type 2 diabet. Their projections expect more than 470 million diagnosed prediabetics people by 2030 [2].

Without intervention, up to 70% of prediabetic individuals are expected to develop the disease within a four-year window. In this context, finding and improving prediabetic and diabetic detection methods becomes a key step to counter this growing phenomenon.

Currently, the clinical standard for diagnosis relies on static metrics such as fasting glucose, oral glucose tolerance tests, and glycosylated hemoglobin (HbA1c). While average-based measurements are widely used,

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glycemic variability is increasingly recognized as a superior metric for predicting cardiovascular risks and identifying metabolic dysfunction [3].

The outlined reasons clearly demonstrate the importance of detecting, monitoring and classifying the different stages of this condition, related with this matter could be consulted for example [4] and [5]. In this regard, these contributions could improve the health of individuals with T2DM and address fundamental public health issues, as well as the prevention of various associated diseases. A recent and interesting report of the relevance of the early detection of this pathology related with the public health can be found in [6], and a for an overview in the same way of research can be consulted [7].

Because glucose levels are regulated by a multi-organ physiological response, the temporal evolution of blood glucose exhibits a chaotic dynamic [8]. Analyzing the full shape of continuous blood glucose curves allows to evaluate the magnitude, frequency, and rate of change of fluctuations.

Recent literature includes several studies dedicated to the detection of DBT2 using machine learning algorithms as can be seen for example in [9] and [10] and [11], and to evaluate their effectiveness, can be consulted for instance [12].

Studies reporting on advances in the detection of this condition through the use of generative artificial intelligence (GAI) are much more recent, see in case of [13] and adding the Internet of Things in the detection process to the GAI are presented among other works can be found in [14].

The main goal of this research was to develop a machine learning model based exclusively on continuous interstitial glucose measurements. To achieve this, continuous glucose monitoring (CGM) time series were analyzed. Two fundamental characteristics were sought: the feature space should be of the lowest possible dimension while still having the capacity to explain disease dynamics.

To achieve these objectives, information-theoretic measures were utilized, specifically permutation entropy and statistical complexity. Building on the findings of a previous study [15], we established that the entropy-complexity causal plane is one of the most appropriate feature space for capturing the underlying dynamics of these signals.

The data analyzed in this work were sourced from three independent databases: an in-house dataset, an MIT-curated gold standard, and a Stanford University study. Because the metrics demonstrated remarkable consistency regardless of the data origin, we were then able to merge these sources into an expanded, unified database. This integration allowed us to formulate a robust classification model that achieves excellent quality parameters during both the training and evaluation stages.

The remainder of this research is organized as follows: Section 2 details the computational tools and databases used to create the optimal training space. Section 3 presents the parameter optimization results for this feature space. Section 4 explores the selection and fine-tuning of the machine learning model. Section 5 outlines how the invariance of the metrics to the signal source supports model validation on the unified dataset. Finally, Section 6 evaluates the classification model's performance across the curated databases and discusses the biological insights gained from this study.

2. Methods and Databases

Continuous Glucose Monitoring (CGM) is a widely used method for studying glucose dynamics in diabetes. By measuring blood glucose levels in interstitial sensor in real-time, CGM provides a minimally invasive approach to track glycemic regulation. Thus, CGM measurements generate continuous time-series data [16].

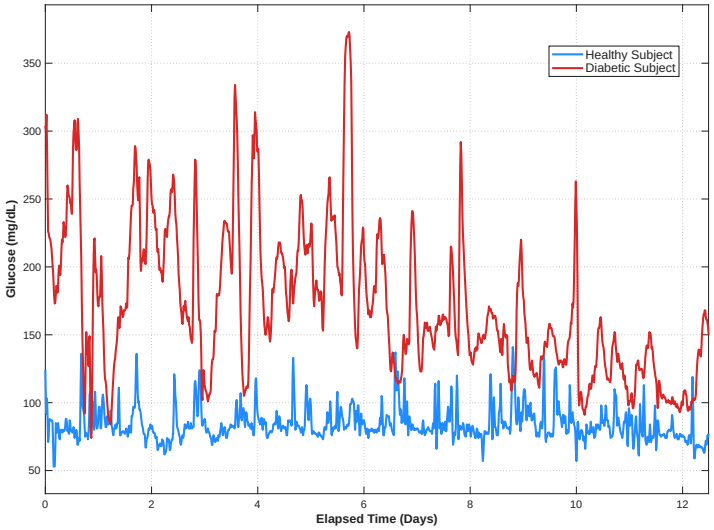


Figure 1. Example of Continuous Glucose Monitoring traces
of diabetic and non diabetic subjects

2.1. Methods

A wide variety of efforts aimed at the effective detection of type 2 diabetes mellitus (T2DM) based on continuous glucose monitoring measurements have been undertaken, just for examples can be consulted [17] and [18] and internal references. In this study, we selected informational measures — primarily entropy and complexity — based on previous results [15], as well as their ability to characterise the dynamics of the system that generates them. It was delved deeper into an aspect related to entropy by studying the benefits of incorporating an entropy variant sensitive to signal amplitude, given the characteristic variations between the two study groups (see Figure 1). To this end, it was proposed the weighted permutation entropy, which will be presented in the following paragraphs. In addition, the values of the parameters needed for the calculations were investigated, using information from the existing literature as well as the specific characteristics of the databases that were used. Next, the databases used in this research are presented below containing the CGM records of both clases of interest.

2.1.1. Entropy

Because glucose dynamics can exhibit chaotic behavior, quantifying their complexity through entropy measures enables precise signal characterization. However, regular metrics such as Shannon Entropy [19] requires long time-series datasets to yield reliable results, making them difficult to apply in real-time physiological monitoring. To address this limitation, Bandt and Pompe (2002) [20] introduced *permutation entropy* (PE), a simple, robust, and computationally efficient complexity measure designed.

By deriving symbolic dynamics directly from relative temporal order rather than arbitrary amplitude binning, permutation entropy offers distinct advatanges. The metric is invariant under strictly monotonic nonlinear transformations while remaining robust under observational and dynamical noise, where conventional parameters fail [21]. Computed rapidly without heavy preprocessing or parameter tuning, PE captures underlying complexity, providing a practical framework for continuous, real-time glucose characterization.

Let $X(t) = \{x_t\}_{t=1...N}$ be a time series of N observations of the variable X. PE essentially measures the occurence of permutation pattern of the ranks of values in $X(t)$. We note π_i the i_{th} possible permutation pattern of $X(t)$. This pattern is caracterized by its length m, called *embedding dimension*, and its sample frequency τ, also named *embedding time delay*.

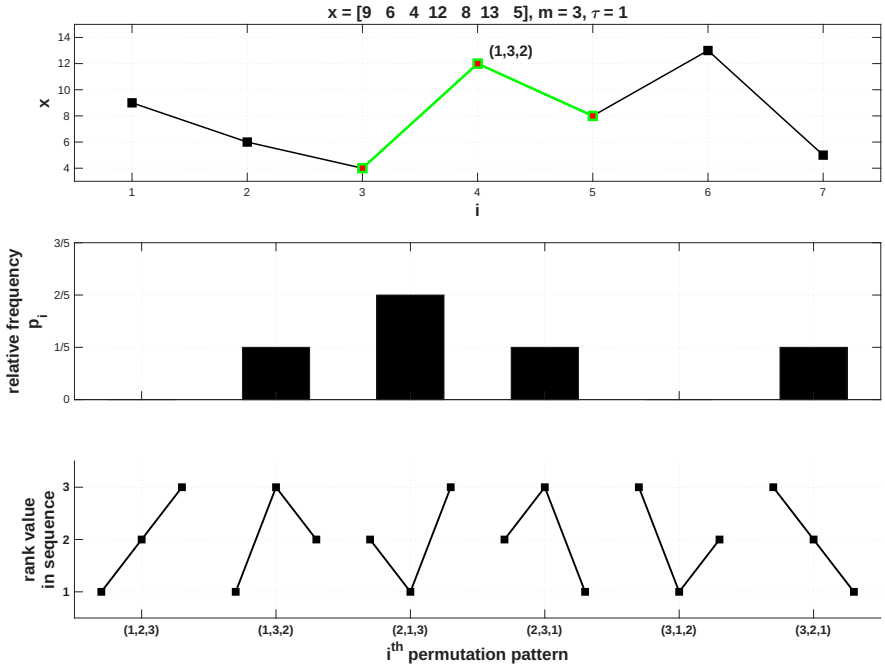


Figure 2. Permutation entropy encoding ($m = 3, \tau = 1$): Sample time series sequence mapped to pattern (1,3,2) (top), full set of possible $n = 4$ patterns (middle), and their relative frequencies for this series (bottom).

Consider the time-delay reprensentation (sample window) $X_j^{m,\tau} = \{x_j, x_{j+\tau}, \ldots, x_{j+(m-1)\tau}\}$. Then let $P = \{p_i\}_{i=1\ldots m!}$ the probability density function (pdf) of the possible permutations $\Pi = \{\pi_i\}_{i=1,\ldots,m!}$, meaning p_i represent the relative frequency of the π_i permutation pattern. Formally, it is defined as

$$p_i = p(\pi_i) = \frac{||\{j : j < m!, type(X_j^{m,\tau}) = \pi_i\}||}{m!}, \tag{1}$$

where $type(.)$ denotes the map from pattern space to symbol space and $||.||$ stands for the cardinality of the set. An alternative way of defining permutation pattern probability is

$$p(\pi_i) = \frac{\sum_{j \leq m!} \mathbb{1}_{u:type(u)=\pi_i}(X_j^{m,\tau})}{\sum_{j \leq m!} \mathbb{1}_{u:type(u)\in\Pi}(X_j^{m,\tau})}, \tag{2}$$

where $\mathbb{1}_A(u)$ denotes the indicator function of the set A.

Band and Pompe defined PE in the same way as Shannon Entropy

$$S(P)_m^\tau = -\sum_{i=1}^{m!} p_i ln(p_i). \tag{3}$$

Let Pe be the uniform distribution pdf. For the rest of the study, the normalized permutation entropy will be used, defined as

$$H = \frac{S[P]}{S[P_e]} = -\frac{1}{m!}\sum_{i=1}^{m!} p_i ln(p_i), \tag{4}$$

and satisfies $0 \leq H \leq 1$. Alternative definitions might be using log_{10} or log_2, changing the units of the results.

2.1.2. Complexity

Then, the definition of complexity given by R. Lopez-Ruiz et al. [22], denoted at C, gives a physical understanding to provide further biological interpretations.

Let Q be the disequilibrium defined as

$$Q[P] = Q_0 D[P, P_e] \tag{5}$$

where Q_0 is a normalization constant and D a stochastic distance. In this work, D is considered as the Jensen-Shannon divergence given by

$$D_{JS}[P, P_e] = S[\frac{P + P_e}{2}] - \frac{1}{2}S[P] - \frac{1}{2}ln(m!), \tag{6}$$

for which

$$Q_0 = -2[\frac{m! + 1}{m!}ln(m! + 1) - 2ln(2m!) + ln(m!)]^{-1}. \tag{7}$$

It can be observed that, as its name suggests, the disequilibrium decreases as the P distribution behaves as the uniform distribution P_e.

Thus, the complexity for a given distribution P can be measured as the combination of inherent entropy and "entropy of pertubation" i.e. disequilibrium

$$C[P] = Q[P]H[P]. \tag{8}$$

2.1.3. Entropy-Complexity Causal Plane

Individually, these two metrics already provide valuable informations on the system behavior: entropy gives good hint whether the studied signal is primarily deterministic, random or chaotic while complexity caracterizes the system's dynamical evolution by tracking how structural organization changes over time.

Combining these metrics forms the Entropy-Complexity Causal Plane, a two-dimensional diagnostic space bounded by theoretical minimum C_{min} and maximum C_{max} complexity curves, for an introduction see [23], and for a recent application in a bioengineering environment [24] . By embedding temporal causality through ordinal patterns, this plane differentiates dynamical behaviors that appear identical under unidimensional analysis, offering a geometric footprint of how the system's underlying biological mechanisms evolve.

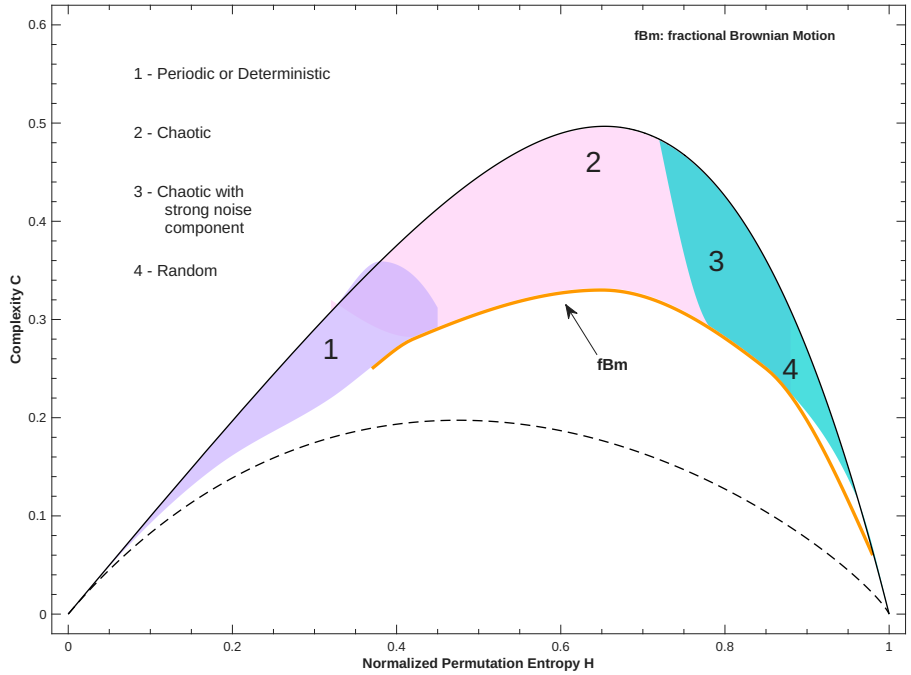


Figure 3. The regions of the Entropy-Causality Plane according to the system behaviour ($m = 5, \tau = 2$)

2.1.4. Weighted Permutation Entropy

As illustrated in Fig. 1, the CGM trace of group 1 exhibits significantly larger fluctuation amplitudes compared to those of group 2. Standard Permutation Entropy (PE) relies solely on the relative rank order of data points, making it invariant to scaling and insensitive to signal magnitude. Consequently, standard PE discards amplitude variations that may contain critical physiological information regarding glycemic variability.

To address this limitation and incorporate signal amplitude into the diagnostic metrics, *Weighted Permutation Entropy* (WPE) can be used [25]. By assigning weights based on the variance or power of each ordinal motif, WPE preserves both the underlying temporal structure and the amplitude dynamics of the signal, enabling a more comprehensive differentiation between the classes of interest.

$$p(\pi_i) = \frac{\sum_{j \leq m!} \mathbb{1}_{u:type(u)=\pi_i}(X_j^{m,\tau}).w_j}{\sum_{j \leq m!} \mathbb{1}_{u:type(u)\in\Pi}(X_j^{m,\tau}).w_j}. \tag{9}$$

One of the method to define the weigths $\{w_j\}$ is by using the variance of each vector $X_j^{m,\tau}$

$$w_j = \frac{1}{m!}\sum_{k=1}^{m}(x_{j+(k-1)\tau} - \overline{X_j^{m,\tau}}), \tag{10}$$

where $\overline{X_j^{m,\tau}}$ is the arithmetic mean of $X_j^{m,\tau}$

$$\overline{X_j^{m,\tau}} = \frac{1}{m}\sum_{k=1}^{m} x_{j+(k-1)\tau}. \tag{11}$$

Then the WPE and its normalized version, denoted as W, are defined the same way as 3 and 4. This ponderated metric of the permutation entropy addresses abrupt changes in magnitude of the signal, oftenly highlighting a change of behavior.

2.2. Databases

During all this study, diabetic subjects will be denominated as group 1 and non diabetic subjects as group 2. The collection of CGM time series studied in this work was obtained from 3 sources:

- A CPSI data collection campaign made in collaboration with argentinian hospitals. This dataset contains 39 records, 19 for group 1 and 20 for group 2 (16 records for each group after data curation). With a sampling period of 15 minutes, it reaches an average of 1200 CGM data points per signal. The two groups of this dataset are noted `DBT2/NDBT2`, for details about this database see [26].

- `CGMacros`, a dataset obtained from the PhysioNet plaform managed by members of the Computational Physiology Laboratory of the Massachusetts Institute of Technology [27]. CGMacros contains data from 45 American participants: 14 for group 1, 15 for group 2 and the rest qualified as prediabetic subjects. After the initial screening, an Abbott FreeStyle Libre Pro CGM (15-min sampling period) and a Dexcom G6 Pro CGM (5-min sampling, resampled to 15-min) were used by physicians leading to CGMacros_dexcom and CGMacros_libre sub-datasets. The curation has led to 12 group 1 and 14 group 2 subjects for CGMacros_dexcom and 13 group 1 and 11 group 2 subjects for CGMacros_libre, with average 1000 points for both groups.

- A study from the Stanford University [8] who have recruited 57 participants without prior diagnosis. In the screening tests, 5 met the criteria of the American Diabetes Association for being diabetic, 14 had prediabetes, and the remainder were non diabetic. Curation leaded to the `Hall` dataset, made of 4 subjetcs for group 1 and 31 for group 2. After decimation, each time series is composed of average 650 data points.

The data used in this study were collected from unpaid volunteers whose anonymity and willingness to participate have been guaranteed. Their data were recorded in accordance with all the guidelines of the Declaration of Helsinki.

To address measurement artifacts present across datasets, an outlier detection criterion was established based on distance to the group centroid. For each dataset, the distance of every data point from its corresponding centroid were computed. A data point is classified as an outlier if its centroid distance exceeded the standard deviation of the group's overall distance distribution. The complete listing of outliers can be found in Appendix A. It leads to the followin final composition:

Dataset	Nb of records in group 1	Nb of records in group 2	Average number of data points
`DBT2/NDBT2`	19	20	1200
`CGMacros_dexcom`	12	14	1000
`CGMacros_libre`	13	11	1000
`Hall`	4	31	650

Table 1. Number of records of each of studied datasets

It is important to take into account that the minimal number of data points is directly correlated with the choice of embedding dimension and embedding time delay (m, τ).

The most important artefact in the calculation of permutation entropy is connected to a too small sample size i.e. N too small compare to the number of possible permutations $m!$. Therefore, litterature advises to follow the recommendation in to choose the maximum m according to $N >> m!$ [28].

Regarding the embedding time delay selection, experience of experts [28] as well as autocorrelation study [29] shows that keeping the smallest values works the best for biomedical time series.

3. The entropy-complexity plane

It is important to remember that this article has has two main objectives. Firstly, it aims to find a reliable way of using the different cited metrics to distinguish between diabetic and non-diabetic subjects, while keeping the number of necessary parameters to a minimum. Furthermore, since permutation entropy-related metrics are directly correlated with physical behaviour [30], another goal of this rearch is to enhance the knowledge about the dynamics of DMT2 pathology.

The objectives of this article are manifold. It aims to find a trustful way of using the differend cited metrics to distinguish diabetics and non-diabetics subjects while always trying to keep the minimum number of necessary parameters to achieve this task. Moreover, as permutation entropy-related metrics are directly correlated with physical behavior, the other goal of this article is to retrospectively enhance the clinical knowledge of mellitus type 2 diabet.

As their definitions show, PE and WPE rely on the choice of m and τ; the first part of this study will be to find the couple of parameters that provide the best separation between the two groups. Once this separation was done, the appliance of the right algorithm permits to optimize the classification of the subjects and, in a third part, to conclude to the performances of the Entropy-Complexity causal plane to detect diabets mellitus type 2.

3.1. Selection of the permutation entropy parameters

One of the approach to find relevant embedding dimension m and embedding time delay τ parameters, in the scope defined in 2.2, is by using barycenters of each cloud of points. As it is the most reliable, the DBT2/NDBT2 dataset was used.

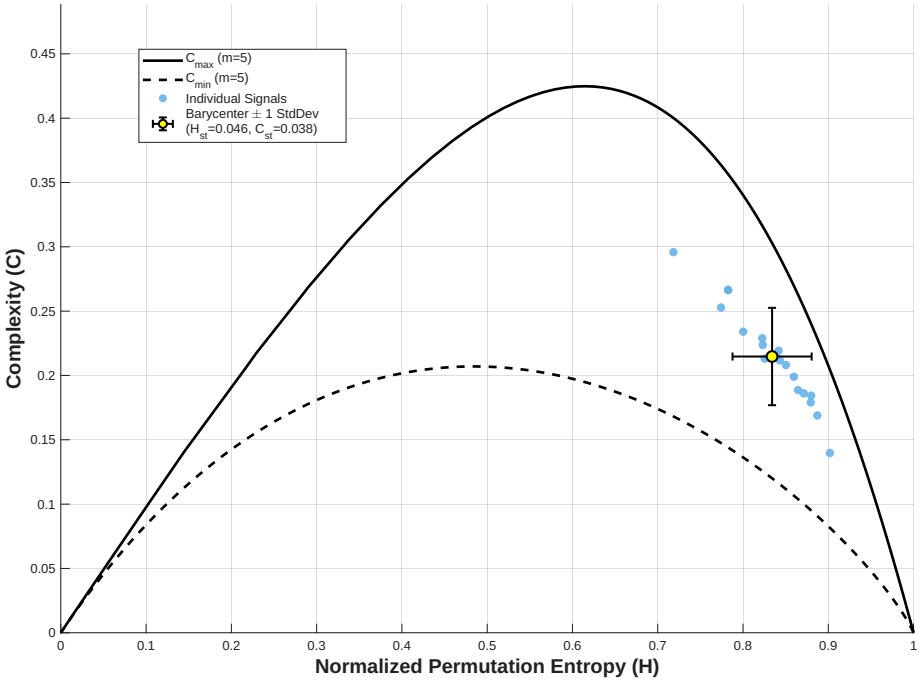


Figure 4. Distribution of DBT2 dataset for $m = 5, \tau = 2$

Then the euclidean distance between the two barycenters becomes a simple criteria to quantify the efficiency of the parameters.

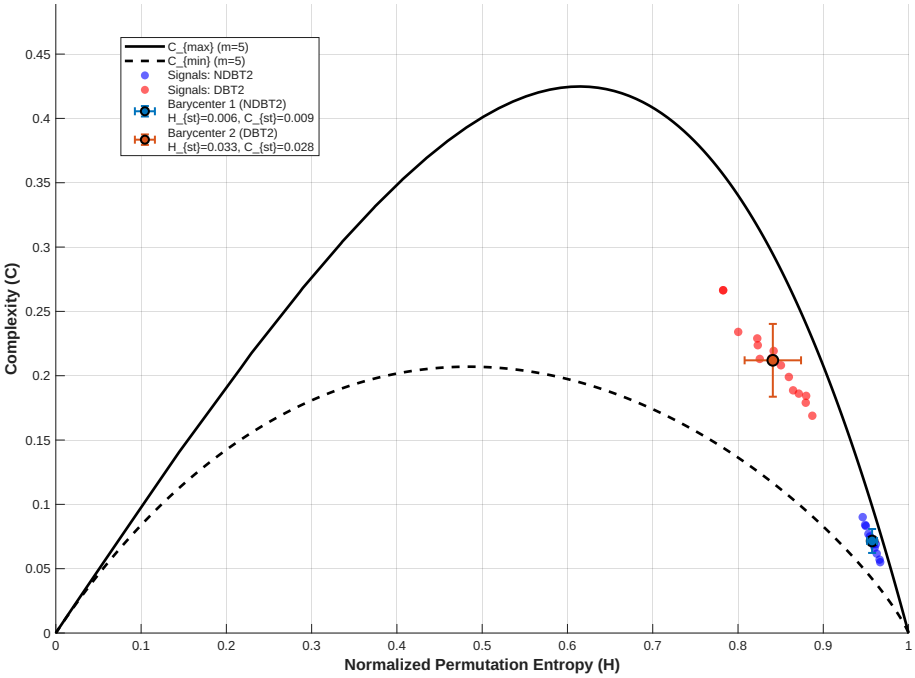


Figure 5. Barycenters comparison between DBT2 and NDBT2 datasets ($m = 5, \tau = 2$)

First thing to recall is that permutation entropy apply while $N >> m!$ with N the size of the sample. Here, datasets allow to test up to $m = 5$. Moreover, the literature suggests to keep τ as low as possible, leading to testing upon to 3.

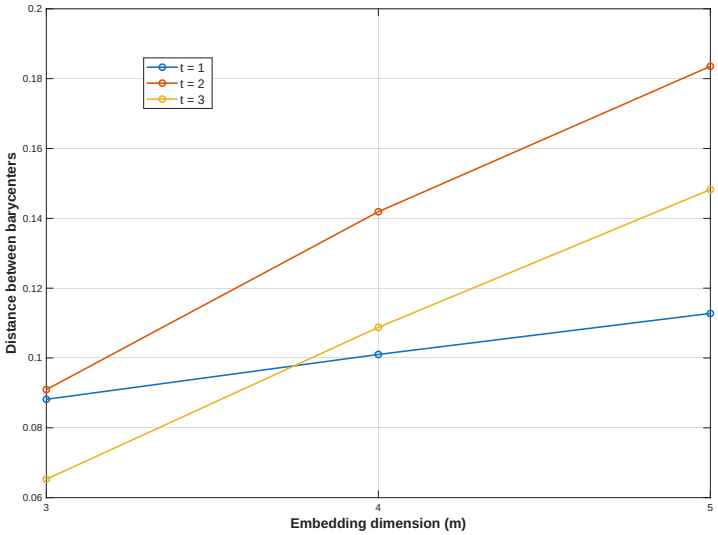


Figure 6. Distance between barycenters vs m for each t computed with standard PE

Thus, the couple $(m = 5, \tau = 2)$ fits the most for computing PE on this type of datas. The same work has been done with weighted PE:

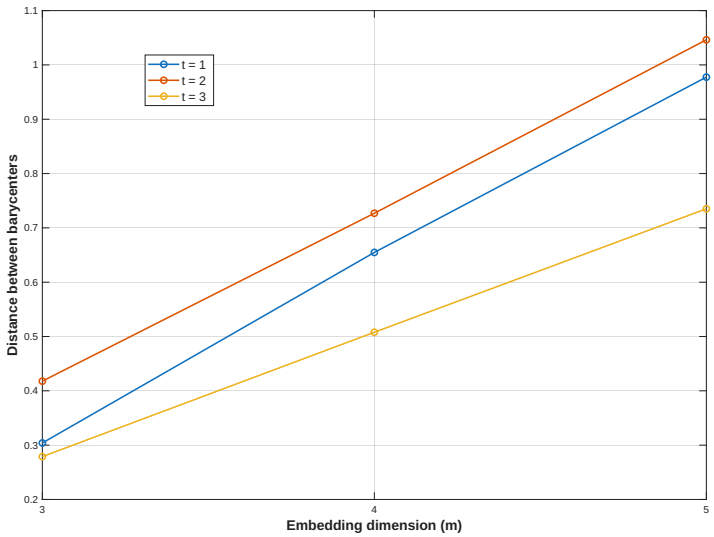


Figure 7. Distance between barycenters vs m for each t computed with WPE

Once again, $(m = 5, \tau = 2)$ gives the best result. Nevertheless, it has been observed that for WPE, using this set of parameters introduce extra shift to the left and spread for every cloud of point:

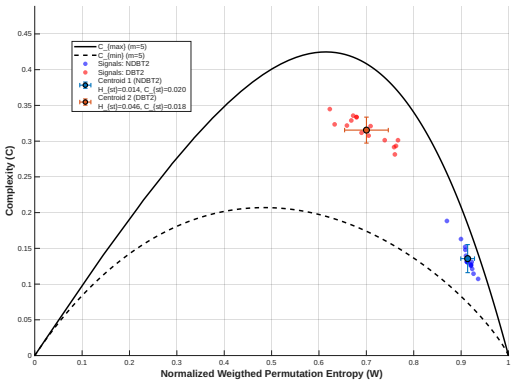


Figure 8. WPE-Complexity causal plane for $(m = 5, \tau = 2)$

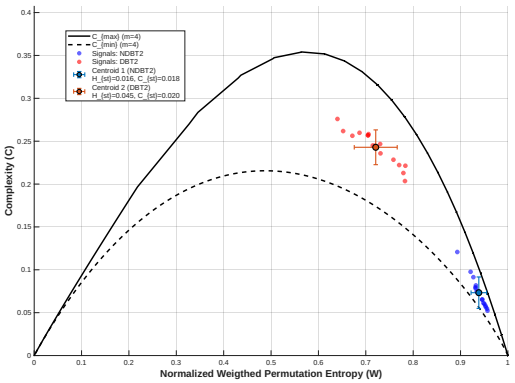


Figure 9. WPE-Complexity causal plane for $(m = 4, \tau = 2)$

Overall, this selection provides good candidates of set of parameters for both PE and WPE that will be used for the rest of the classification process.

Entropy Definition	Best (m, τ)
Permutation Entropy	(5,2)
Weighted Permutation Entropy	(4,2)

Table 2. Optimized sets of parameters for each information metric

4. Classification process

4.1. Optimizing classification algorithm

The great separation and agregation of each cloud of points on the DBT2/NDBT2 dataset, provided by the use of the right parameters (see Fig. 2), is a great start to apply clustering algorithms such as k-means.

The k-means algorithm is an unsupervised machine learning technique designed to partition n observations into k distinct, non overlapping clusters [31]. In this study, $k = 2$ to mirror the group 1 and group 2 populations. Mathematically, the algorithm minimizes the Within-Cluster Sum of Squares (WCSS) by iteratively executing two steps until convergence: assigning each data point to the cluster governed by the nearest Euclidean centroid, and subsequently updating each centroid to the spatial mean of its assigned points [32].

A well-known limitation of k-means is its sensitivity to initial centroid selection, which can cause the algorithm to become trapped in local minima rather than achieving global optimization. However, the parameter optimization applied during feature extraction (PE and WPE) directly counters this drawback. By maximizing class dispersion while maintaining compact intra-class variance, the enhanced cluster separability stabilizes the optimization landscape and helps the algorithm to reach a great global partition, regardless of the initialization.

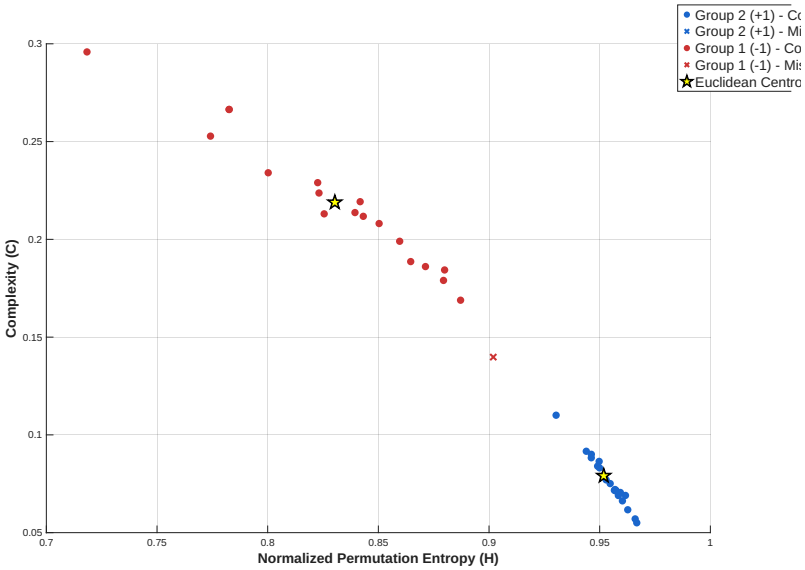


Figure 10. K-means algorithm applied on the Entropy-Complexity causal plane with the uncurated DBT2/NDBT2 dataset

Without any optimization of the algorithm, this method ensures very good classification results, even on uncurated dataset.

4.2. Classification performances

To evaluate and optimize k-means MATLAB's implementation performance, several hyperparameter configurations were investigated. First, spatial separation between data points and centroids was computed using four distinct distance metrics: squared Euclidean (*sqeuclidean*), Manhattan (*cityblock*), cosine distance (*cosine*), and correlation distance (*correlation*). Convergence limits were set via the maximum iteration parameter (*MaxIter* $\in \{100, 1000, 5000\}$), capping the allowable assignment and update loops. To mitigate k-means' sensitivity to initial conditions and prevent trapping in suboptimal local minima, the algorithm was executed across multiple replicates ($T \leq 5$), running the full optimization T times with newly generated initial centroids and retaining the solution with the minimum total variance. Finally, three centroid initialization strategies (*plus* with k-means++, *sample*, and *cluster*) were evaluated to dictate how the starting k centroids were selected prior to the first iteration.

The *plus* strategy implements the "k-means++" algorithm, a probabilistic seeding technique engineered to choose spatially dispersed initial centroids across the feature domain. After selecting an initial centroid c_1 uniformly at random from dataset X, each subsequent centroid is drawn with a probability proportional to the squared distance $D(x)^2$ between candidate point x and its nearest already-selected centroid, such that:

$$P(x) = \frac{D(x)^2}{\sum_{x_i \in X} D(x_i)^2} \tag{12}$$

By penalizing candidate centers near established centroids, k-means++ prevents seed crowding in dense data regions, significantly accelerating iteration convergence while providing theoretical upper bounds ($O(\log k)$) on clustering error relative to the global optimum.

In contrast, the *sample* strategy serves as a uniform random sampling baseline wherein k unique observations are selected without replacement directly from dataset X to serve as initial centroid coordinates. Although this approach incurs zero computational overhead during initialization, it is entirely agnostic to spatial density distributions. Consequently, randomly selected observations frequently cluster in high-density regions or capture

extreme outliers, increasing the number of assignment-update iterations required to stabilize and elevating the risk of termination in poor local minima.

Finally, the *cluster* strategy employs a two-stage heuristic designed to discover optimized seed locations for large-scale datasets. A preliminary k-means pass, initialized via uniform sampling, is performed on a randomly extracted subsample (typically 10%) of the primary dataset. The k centroid coordinates derived from this fast sub-run are then extracted and supplied as the starting locations for the primary k-means optimization on the complete dataset. By identifying coarse structural topology on a reduced volume of data, this pre-clustering scheme provides representative starting positions that substantially reduce the full-dataset iteration count without incurring the distance-calculation overhead of k-means++ across large sample sizes.

To evaluate classifier performance and assess variable contributions, the following evaluation metrics were computed:

- **Balanced Accuracy (BA):** Measures overall diagnostic accuracy across potentially imbalanced classes by taking the arithmetic mean of Sensitivity and Specificity:

$$BA = \frac{1}{2}\left(\frac{TP}{P} + \frac{TN}{N}\right). \tag{13}$$

- **F1-score (F_1):** Computes the harmonic mean of precision and recall (Sensitivity), providing a robust measure of predictive performance in the presence of class imbalance:

$$F_1 = \frac{2 \cdot TP}{2 \cdot TP + FP + FN}. \tag{14}$$

- **Sensitivity (True Positive Rate, TPR):** Quantifies the proportion of actual positive cases correctly identified:

$$\text{Sensitivity} = \frac{TP}{P} = \frac{TP}{TP + FN}. \tag{15}$$

- **Specificity (True Negative Rate, TNR):** Quantifies the proportion of actual negative cases correctly identified:

$$\text{Specificity} = \frac{TN}{N} = \frac{TN}{TN + FP}. \tag{16}$$

where TP, TN, FP, and FN represent true positives, true negatives, false positives, and false negatives, respectively, while P and N denote total positive and negative instances.

Finding the best configuration for each metric has then been done by computing the 240 possible combinations of parameters (4 distance metrics $\times$ 3 iteration limits $\times$ 5 replicate counts $\times$ 3 initialization strategies). Because several distinct configurations yielded identical top-tier performance, Table 3 reports a representative optimal configuration for each metric.

Metric	Distance	MaxIter	Replicates	Start	Balanced Accuracy	F1-score	Sensitivity	Specificity
Best Balanced Accuracy	`cosine`	100	1	`plus`	**0.93**	0.92	0.92	0.93
Best F1-score	`sqeuclidean`	1000	1	`plus`	0.93	**0.93**	0.92	0.94
Best Sensitivity	`cosine`	5000	1	`plus`	0.93	0.93	**0.92**	0.94
Best Specificity	`correlation`	100	1	`plus`	0.51	0.04	0.02	**1.00**

Table 3. Example of top configurations by metric for the k-means algorithm applied on the DBT2/NDBT2 dataset

Then the search of configurations that match the maximum number of best metrics at the same time retains the best candidates

	Best Configuration				Quality Metrics			
Entropy Definition	**Dist.**	**MaxIter**	**Repl.**	**Start**	**Acc.**	**F1**	**Sens.**	**Spec.**
Permutation Entropy ($m = 5, \tau = 2$)	`cosine`	100	1	`plus`	0.93	0.93	0.92	0.94
Weighted Permutation Entropy ($m = 4, \tau = 2$)	`sqeuclidean`	100	1	`plus`	0.97	0.98	1.00	0.95

Table 4. Best k-means configuration depending on entropy definition.

It is important to keep in mind that, thanks to the preliminary separation of the cloud points, around 100 combinations of parameters lead to the best quality metrics for each version of the permutation entropy. Keeping those result of the choice of keeping computation as simple and fast as possible.

4.2.1. Within-Datasets

Applying the k-means optimized algorithm on the Permutation Entropy-Complexity causal plane led to the following performances:

Target group	Balanced Accuracy	F1-score	Sensitivity	Specificity
DBT2/NDBT2	0.98	0.98	0.95	0.91
CGMacros_dexcom	0.79	0.81	0.87	0.71
CGMacros_libre	0.90	0.90	0.87	0.93
DBT2/NDBT2 + CGMacros	0.93	0.93	0.92	0.94
Hall	0.85	0.93	0.89	0.80

Table 5. Quality metrics within dataset for PE-Complexity causal plane

While keeping the same algorithm, switching to the Weighted Permutation Entropy and its correlated definition of complexity allowed improved results:

Target group	Balanced Accuracy	F1-score	Sensitivity	Specificity
DBT2/NDBT2	1.00	1.00	1.00	1.00
CGMacros_dexcom	0.96	0.96	0.92	1.00
CGMacros_libre	1.00	1.00	1.00	1.00
DBT2/NDBT2 + CGMacros	0.99	0.99	0.97	1.00
Hall	0.88	0.98	1.00	0.75

Table 6. Quality metrics within dataset for WPE-Complexity causal plane

As detailed in Tables 5 and 6, incorporating signal amplitude via WPE tends to outperform standard PE across all target groups. Under the WPE-Complexity plane, classification performance reaches near-perfect accuracy, achieving a Balanced Accuracy of 1.00 for both `DBT2/NDBT2` and `CGMacros_libre`. Across both entropy metrics, `CGMacros_libre` consistently yields higher metrics than `CGMacros_dexcom` (e.g., 0.90 vs. 0.79 Balanced Accuracy for PE; 1.00 vs. 0.96 for WPE), highlighting potential differences in sensor dynamics or signal-to-noise ratio between devices.

4.2.2. Cross-Datasets

Checking clusters reusability is a common way to further performances' characterization of the k-means algorithm. To be done, the coordinates of the clusters of a first dataset, called evaluation context, are used as a "nearest-neighbor" classifier, using the traditional squared Euclidean distance.

Target group	Evaluation Context	Balanced Accuracy	F1-score	Sensitivity	Specificity
CGMacros_dexcom	DBT2/NDBT2	0.83	0.81	0.73	0.93
CGMacros_libre	DBT2/NDBT2	0.93	0.93	0.87	1.00
Hall	DBT2/NDBT2 + CGMacros	0.86	0.95	0.92	0.80

Table 7. Quality metrics cross dataset for PE-Complexity feature set

One more time, these results can be compared with WPE's ones.

Target group	Evaluation Context	Balanced Accuracy	F1-score	Sensitivity	Specificity
CGMacros_dexcom	DBT2/NDBT2	0.96	0.96	0.92	1.00
CGMacros_libre	DBT2/NDBT2	1.00	1.00	1.00	1.00
Hall	DBT2/NDBT2 + CGMacros	0.88	0.98	1.00	0.75

Table 8. Quality metrics cross dataset for WPE-Complexity feature set

Tables 7 and 8 show that the extracted features maintain high cross-dataset transferability with negligible loss in predictive performance. The WPE feature plane shows promising generalizability, supporting reliable sample separation within the experimental setup (1.00 Balanced Accuracy, F1-score, Sensitivity, and Specificity) when testing `CGMacros_libre` on the `DBT2/NDBT2` context. Evaluation on the `Hall` dataset exhibits perfect Sensitivity (1.00 under WPE) alongside a relatively lower Specificity (0.75–0.80), pointing to minor class imbalance or higher intra-group variance in that cohort. Overall, the WPE-Complexity plane can provide a new feature set to be used in machine learning algorithm.

5. Construction of the whole database

The decision to merge the individual datasets into a single, unified database was fundamentally driven by the behavior of the continuous glucose monitoring time series when projected onto the information-theory domain. Despite the inherent heterogeneity in sensor technologies, sampling rates, and demographic cohorts across the separate studies, preliminary analysis revealed a striking geometrical overlap in their WPE and complexity metrics.

Because WPE and associated complexity capture the intrinsic non-linear dynamics and structural temporal patterns of the signals rather than absolute amplitude fluctuations, they are highly resistant to scale, device-specific artifacts, and varying baseline levels. When mapped onto the WPE-Complexity causal plane, the coordinate distributions for diabetic and non-diabetic subjects from the distinct databases exhibited significant spatial overlap. The dynamical features of the signals clustered according to the underlying physiological condition rather than the origin of the dataset. This topological consistency demonstrated that the entropy-complexity footprint of glucose dynamics is a universal physiological marker, fully justifying the concatenation of the records to form a larger, more robust database for training and evaluating the clustering algorithm.

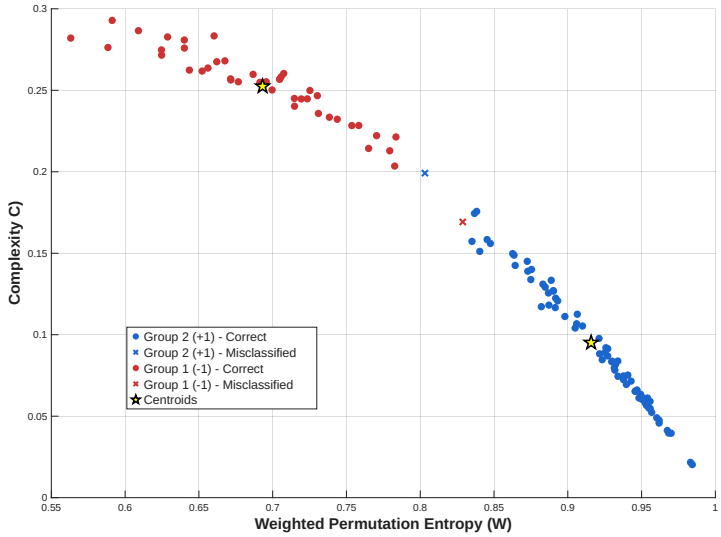


Figure 11. K-means algorithm applied on the WPE-Complexity causale plane with the mix of all datasets

As illustrated in Figure 11, applying the optimized k-means algorithm to the unified dataset—aggregating all curated records from the `DBT2/NDBT2`, `CGMacros`, and `Hall` databases yields exceptional diagnostic performance on the WPE-Complexity causal plane. The classifier achieves a global Accuracy of 98.1% and an F1-score of 0.99, demonstrating highly robust predictive capabilities despite the underlying heterogeneity in sensor types, sampling frequencies, and demographic cohorts. Furthermore, the model exhibits a Sensitivity of 0.99 and a Specificity of 0.98, indicating its reliability in correctly identifying diabetic subjects while maintaining an extremely low false-positive rate for non-diabetic individuals. This cohesive clustering confirms that the temporal and amplitude dynamics of continuous glucose monitoring, when projected onto the weighted feature space, could offer a universal and highly separable geometric footprint for type 2 diabetes.

6. Conclusion

The two main objectives set out at the start of this research have been successfully achieved. It has been demonstrated that a feature space can be formulated with just two variables: weighted permutation entropy and associated statistical complexity. These variables are calculated from a simple, minimally invasive blood glucose concentration measurement using continuous glucose monitors (CGM). The use of permutation entropy, combined with complexity on a causal plane, has proven to be a promising tool for diagnosing type 2 diabetes mellitus (T2DM).

Furthermore, a third result has been obtained in the form of a methodology that could potentially be used to merge heterogeneous databases from different sources. Because the three databases exhibited the same behavior in the entropy-complexity plane when characterizing both classes of interest, it was possible to construct a comprehensive database that contributed to the formulation of more reliable models. While this approach requires a preliminary calibration phase to select the best (m, τ) combination, it creates a favorable ground for k-means clustering. When fine-tuned, the k-means algorithm provides very good classification results, which are further enhanced by taking into account the amplitude dynamic of the CGM using WPE.

Both PE and WPE directly reflect the underlying physiological dynamics of blood glucose regulation. Analysis of the study groups (subjects with and without T2DM) using these informational tools shows a clear differentiation. This points towards a trend observed in previous research: the effect of the condition on the glucose homeostasis

system reduces the entropy value of the signal, decreasing its variability and indicating a loss of regulatory capacity in the physiological system. At the same time, an increase in statistical complexity indicates the presence of a greater number of signal structures within the system that may be linked to the same process. As highlighted in Figs. 3 and 5, the cloud of points corresponding to group 1 exhibits a distinct shift toward lower normalized entropy and higher statistical complexity, indicative of more structured, deterministic chaotic behavior. In contrast, group 2 trajectories display more stochastic behavior.

This dynamical signature offers a novel, complementary framework for characterizing type 2 diabetes. The restricted and structured diabetic dynamics, formalized as higher complexity in Eq. (8), provide a additional diagnostic tool. Ultimately, this methodology leads to efficient and frugal decision support in the diagnosis of type 2 diabetes.

The strengths of this research include the conceptual clarity of the parameters that constitute the feature space, the simplicity of its implementation, its computational efficiency, its low sensitivity to noise and the clarity with which the results can be interpreted. Following further studies on a broader scale and validation by specialists, these characteristics make the future transfer of this research to clinical practice feasible to assist in the diagnosis of the condition.

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7. Appendix

A. Excluded outliers for each datasets

Dataset	Count	Outlier File Identifiers
DBT2	3	resultado_DBT2_sujeto1.mat, resultado_DBT2_sujeto5.mat, resultado_DBT2_sujeto12.mat
NDBT2	4	resultado_DBT2_sujeto2.mat, resultado_DBT2_sujeto5.mat, resultado_DBT2_sujeto9.mat, resultado_DBT2_sujeto11.mat
CGMacros dexcom DBT2	3	CGMacros-028_dexcom_gl.mat, CGMacros-030_dexcom_gl.mat, CGMacros-042_dexcom_gl.mat
CGMacros dexcom NDBT2	2	CGMacros-002_dexcom_gl.mat, CGMacros-033_dexcom_gl.mat
CGMacros libre DBT2	2	CGMacros-028_dexcom_gl.mat, CGMacros-030_dexcom_gl.mat
CGMacros libre NDBT2	4	CGMacros-001_dexcom_gl.mat, CGMacros-002_dexcom_gl.mat, CGMacros-004_dexcom_gl.mat, CGMacros-033_dexcom_gl.mat
Hall DBT2	2	subject_2133-039_DBT2.mat
Hall NDBT2	7	subject_1636-69-069_NDBT2.mat, subject_1636-70-1002_NDBT2.mat, subject_1636-70-1008_NDBT2.mat, subject_2133-001_NDBT2.mat, subject_2133-013_NDBT2.mat, subject_2133-020_NDBT2.mat, subject_2133-022_NDBT2.mat